

Passive Risk Detection Report

devdaytest1 | Harness CD | Cluster: validation-cluster | Namespace: default

Scan Identity: test-scan-1778747446 | Scanned 2026-05-14 10:18 UTC

HARNESS RESILIENCE TESTING - EXPERIMENTAL | Confidential



Executive Summary

This Passive Risk Detection scan (Validation Test Scan - 2026-05-14 14:00:46) identified 7 risks across 1 service(s) in the 'default' namespace on cluster 'validation-cluster'. A total of 5 High, 2 Medium, and 0 Low severity risks were detected, all currently in Open status. The composite Pipeline Risk Score is 58/1000. Immediate remediation is recommended for all High severity findings to prevent service disruption.

Scan Metrics

Service Count	1
Total Risks	7
Services Affected	nginx
Chaos Coverage	0% (no chaos tests run yet)
Open Risks	7
Scan Type	kubernetes-security

Services Found

Service	Total Risks	High	Medium	Low	RRS
nginx	7	5	2	0	58

RRS (Risk Resilience Score) = High x 10 + Medium x 4 + Low x 1. Lower is better. Maximum possible score per service is 1000.

Risk Detail Cards

HIGH

k8s-missing-resource-limits

Status: Open

The nginx container has no CPU or memory limits. It can consume all available node resources, starving sibling pods.

Affects: Deployment/nginx in namespace hce

Impact: Unbounded resource consumption causes noisy-neighbour failures on the node. Memory hog can trigger kernel OOM kills on sibling pods.

First Detected: 2026-05-14 10:21 UTC

Last Detected: 2026-05-14 10:21 UTC

kubernetes

reliability

nginx

HIGH

k8s-single-replica

Status: Open

The nginx Deployment runs with replicas: 1, making it a single point of failure. A pod-delete fault causes 100% service unavailability until the pod restarts.

Affects: Deployment/nginx in namespace hce

Impact: Complete service downtime during any pod disruption, node maintenance, or OOM kill. No redundancy to absorb pod-level failures.

First Detected: 2026-05-14 10:18 UTC

Last Detected: 2026-05-14 10:18 UTC

kubernetes

availability

nginx

MEDIUM

k8s-missing-resource-requests

Status: Open

The nginx container has no CPU or memory requests. It is assigned BestEffort QoS and will be the first pod evicted under any node pressure.

Affects: Deployment/nginx in namespace hce

Impact: BestEffort QoS means nginx is evicted before any other pod on the node when memory or CPU pressure occurs - even if nginx itself is not causing the pr...

First Detected: 2026-05-14 10:22 UTC

Last Detected: 2026-05-14 10:22 UTC

kubernetes

reliability

nginx

HIGH

k8s-missing-probes

Status: Open

The nginx container has no livenessProbe or readinessProbe. Kubernetes routes traffic to unhealthy pods and never restarts stuck ones.

Affects: Deployment/nginx in namespace hce

Impact: Traffic is sent to pods that are restarting or deadlocked. Without livenessProbe, a hung nginx process is never restarted automatically.

First Detected: 2026-05-14 10:22 UTC

Last Detected: 2026-05-14 10:22 UTC

kubernetes

availability

nginx

HIGH

k8s-missing-network-policy

Status: Open

The nginx Deployment has no NetworkPolicy. All ingress and egress traffic is unrestricted, enabling lateral movement from any compromised pod in the cluster to reach nginx directly.

Affects: Deployment/nginx in namespace hce

Impact: Any pod in any namespace can send traffic to nginx without restriction. A compromised workload can exfiltrate data, probe nginx endpoints, or use it a...

First Detected: 2026-05-14 15:14 UTC

Last Detected: 2026-05-14 15:14 UTC

kubernetes

security

network

nginx

MEDIUM

k8s-missing-pod-anti-affinity

Status: Open

The nginx Deployment has no podAntiAffinity or topologySpreadConstraints. All replicas may be scheduled on the same node, turning the multi-replica deployment into a node-level single point of failure.

Affects: Deployment/nginx in namespace hce

Impact: Despite having replicas configured, all pods could land on the same node. A single node-drain or node-restart takes down every replica simultaneously ...

First Detected: 2026-05-14 10:33 UTC

Last Detected: 2026-05-14 10:33 UTC

kubernetes

availability

nginx

HIGH

k8s-latest-image-tag

Status: Open

The nginx container uses image 'nginx' with no tag. A pod restart may pull a different version from the registry, causing silent version drift between replicas.

Affects: Deployment/nginx in namespace hce

Impact: Non-deterministic deployments - two pods in the same Deployment can silently run different image versions after independent restarts.

First Detected: 2026-05-14 10:19 UTC

Last Detected: 2026-05-14 10:19 UTC

kubernetes

reliability

nginx

Severity Distribution

Severity Level	Count	% of Total	Definition
High	5	71.4%	Critical flaws causing immediate service disruption or complete failure.
Medium	2	28.6%	Significant issues degrading reliability or availability under stress.
Low	0	0.0%	Minor issues that may cause problems under specific edge-case conditions.

Category Breakdown

Category	Count	Risks Included
Availability	3	k8s-single-replica, k8s-missing-probes, k8s-missing-pod-anti-affinity
Reliability	3	k8s-missing-resource-limits, k8s-missing-resource-requests, k8s-latest-image-tag
Security	1	k8s-missing-network-policy

Recommendations & Next Steps

Priority	Service	Action	Experiment Type
P1	nginx	Define CPU/memory limits in container spec	CPU Stress / Memory Hog
P1	nginx	Increase replicas to >= 2 and add PDB	Pod Delete
P2	nginx	Set CPU/memory requests for Guaranteed QoS	Node CPU Hog
P1	nginx	Add livenessProbe and readinessProbe	Pod Delete / HTTP Probe
P1	nginx	Create NetworkPolicy to restrict ingress/egress	Network Loss / Partition
P2	nginx	Add podAntiAffinity or topologySpreadConstraints	Node Drain
P1	nginx	Pin image to a specific SHA or semver tag	Container Kill

RECOMMENDED IMMEDIATE ACTION

The 'nginx' Deployment in namespace 'default' (cluster: validation-cluster) runs with a single replica and no liveness/readiness probes. Execute a Pod Delete chaos experiment immediately: delete the nginx pod and confirm it self-heals within 60 seconds. Until replicas >= 2 and probes are configured, any pod disruption results in 100% downtime.

AUTO-PARAMETERISED EXPERIMENT CONFIG

namespace: default
blast_radius: 1 pod (nginx)
hypothesis: nginx pod recovers within 60s; HTTP 200 restored
duration: 60s
rollback_trigger: HTTP 5xx rate > 10% OR recovery_time > 120s

Test Coverage Gap Analysis

Service	Chaos Tests	Load Tests	Coverage %	Notes
nginx	0	0	0%	Critical
SUMMARY	0	0	0%	No chaos or load tests exist. Immediate action required.

Appendix

This report was generated by the Harness Resilience Testing Passive Risk Detection engine. The engine continuously inspects Kubernetes workload configurations in the specified cluster and namespace, comparing observed deployment manifests against a curated library of resilience anti-patterns. Each detected anti-pattern is recorded as a risk with a severity rating, a textual description, and a structured impact assessment.

Risk scores are computed using a weighted formula: $\text{High} \times 10 + \text{Medium} \times 4 + \text{Low} \times 1$, normalized to a scale of 0-1000. The report is classified EXPERIMENTAL and is intended for internal use by SRE, platform engineering, and chaos engineering teams. All timestamps are in UTC. Epoch values are sourced directly from the scan result JSON and converted to human-readable form. No data has been fabricated or interpolated.

Glossary

Term	Definition
RRS	Risk Resilience Score. Weighted risk score: $\text{High} \times 10 + \text{Medium} \times 4 + \text{Low} \times 1$.
PDB	PodDisruptionBudget. Kubernetes resource limiting voluntary pod disruptions.
HPA	HorizontalPodAutoscaler. Kubernetes resource for automatic pod scaling.
MTTR	Mean Time To Recovery. Average time to restore a service after a failure.
PRD-Report	Passive Risk Detection Report. This document.
HIGH	Severity level indicating critical risk with immediate production impact.
MEDIUM	Severity level indicating significant risk under stress or fault conditions.
LOW	Severity level indicating minor risk with limited blast radius.